COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE031	**SECTION:**	CPE21S1
DATE PERFORMED:	10/10/2026	**DATE SUBMITTED:**	10/10/2026
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ACTIVITY NO. 9 Exploring Data Visually

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1

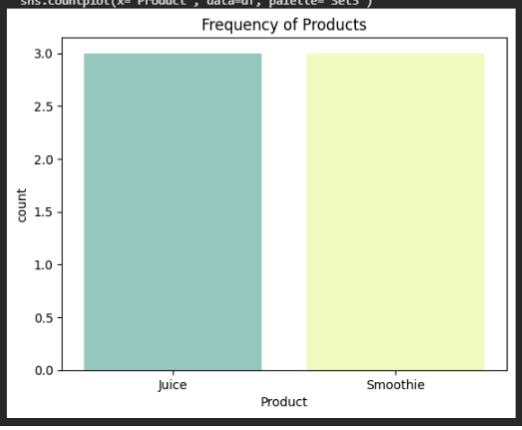
Task:
Create one countplot for Product and one histogram for Profit.
Compare their patterns and interpret what they mean.

```
1 #countplot for product
2 sns.countplot(x="Product", data=df, palette="Set3")
3 plt.title("Frequency of Products")
4 plt.show()
5
```

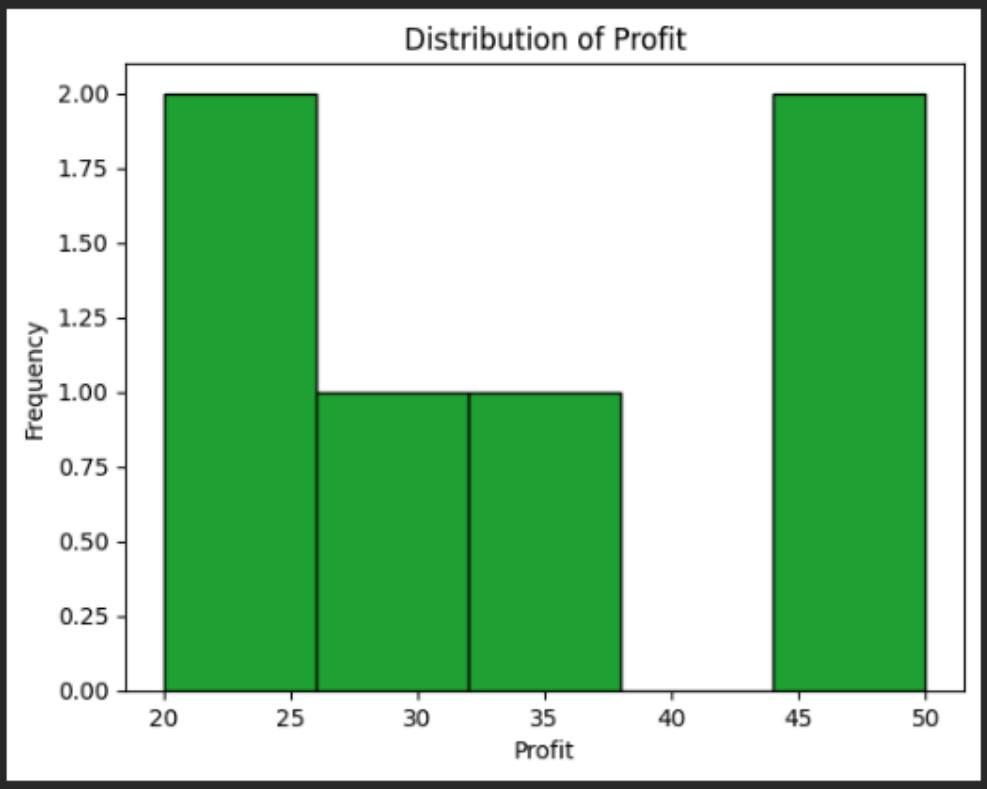
```
/tmp/ipykernel_7960/2222742270.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(x="Product", data=df, palette="Set3")
```



```python
#histogram for profit
plt.hist(df["Profit"], bins=5, color="tab:green", edgecolor="black")
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
```



COMPARISON

For the countplot of products, since the given dataset has equal quantities of Juice and Smoothie, it would naturally reflect in the graph that they are both three. This is different from Profit, it has some variety of numbers. Based on the graph, profit sits around aproximately 20-27 and 44-50. The countplot for products can be interpreted as to what are the regions' most chosen product or what they sell comparing to the other product. For the histogram, I interpret it as to how much do we usually profit and such.

Task 2

Task:
Interpret whether higher sales also mean higher profit.
Which region seems to perform best?

**Based on the scatterplot, higher sales does mean profit. If we look closely at the graph and data table, we can see that East both has the least and the largest profit out of the 4 regions. We take a look back at the data set and see that Smoothie had more sales than Juice in the East. This resulted in more profit in selling Smoothies rather than Juice in the East region.

Task 3

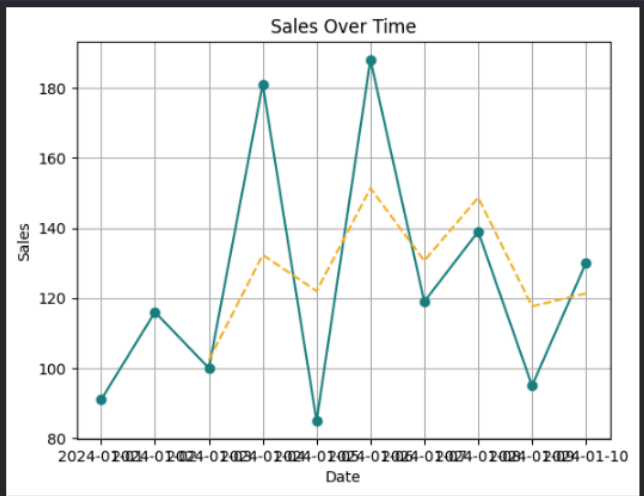
Task:
Describe what you observe in the missing data visualization.
Which variables need attention before analysis?

What I observed in the missing data visualization is that Sales on the 5th region and profit on the 2nd region is shaded red. Before complete analysis, data analyzers should bring their attention to the missing values present by the missing data visualization graph.

Task 4

Task:
Add a moving average line (rolling mean) to smooth fluctuations.
Hint: Use ts_df["Sales"].rolling(window=3).mean().

```
rolling_mean = ts_df["Sales"].rolling(window=3).mean()
plt.plot(ts_df["Date"], ts_df["Sales"], marker="o", color="teal")
plt.plot(ts_df["Date"], rolling_mean, color="orange", linestyle="--")
plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



Task 5
Task: Interpret which branch has the highest sales geographically. How can such visualization help in business decision-making? **Based on the Geospatial Visualization, The South branch has a whopping 150 sales located in the 123x, 8.5y coordinates. These kinds of visualization can help businesses choose which branch needs more resources, attention, marketing, advertising and etc. It clearly presents where the branches are via longitude and latitue degrees and the intensity of darkness depends on the amount of sales in that specific area point.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Create your own mini exploratory analysis:
    - Choose a dataset (e.g., from Kaggle, or a CSV file you have).

2. Perform:
    - Univariate analysis (1 categorical + 1 quantitative)
    - Bivariate analysis (scatterplot or crosstab)
    - Missing data visualization
    - Time-series or geospatial visualization (choose one)

3. Summarize your key insights using visual interpretation.

CODE

```
url = "https://raw.githubusercontent.com/Pasioncpe/CPE-031-CPE21S1---Visualizations-and-Data-Analysis---PASION/refs/heads/main/Anime.csv"

df = pd.read_csv(url)

#categorical
sns.countplot(x="Type", data=df, palette="Set2")
plt.title("Frequency of Media Type")
plt.show()

#quantitative
plt.hist(df["Rating"], bins=16, color="skyblue", edgecolor="black")
plt.title("Distribution of Ratings")
plt.xlabel("Ratings")
plt.ylabel("Frequency")
plt.show()

#bivariate
sns.scatterplot(x="Rating", y="Episodes", data=df, color="coral", s=100)
plt.title("Relationship between Episodes and Ratings")
plt.show()

#missing
df_missing = df.copy()
df_missing.loc[2, "Type"] = np.nan
df_missing.loc[5, "Episodes"] = np.nan
df_missing.loc[5, "Release Season"] = np.nan

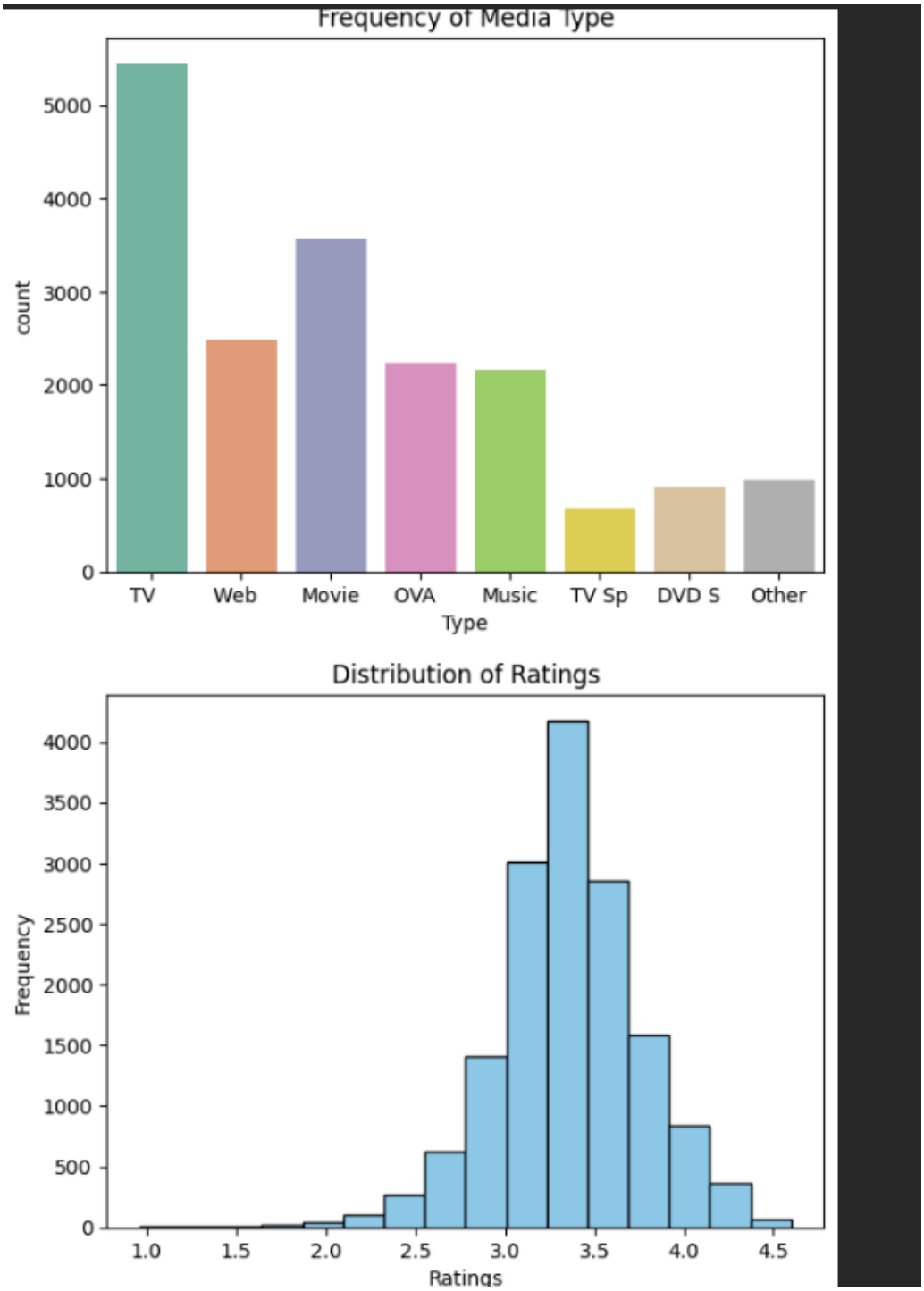
subset_df = df_missing[["Type", "Episodes", "Release Season"]]

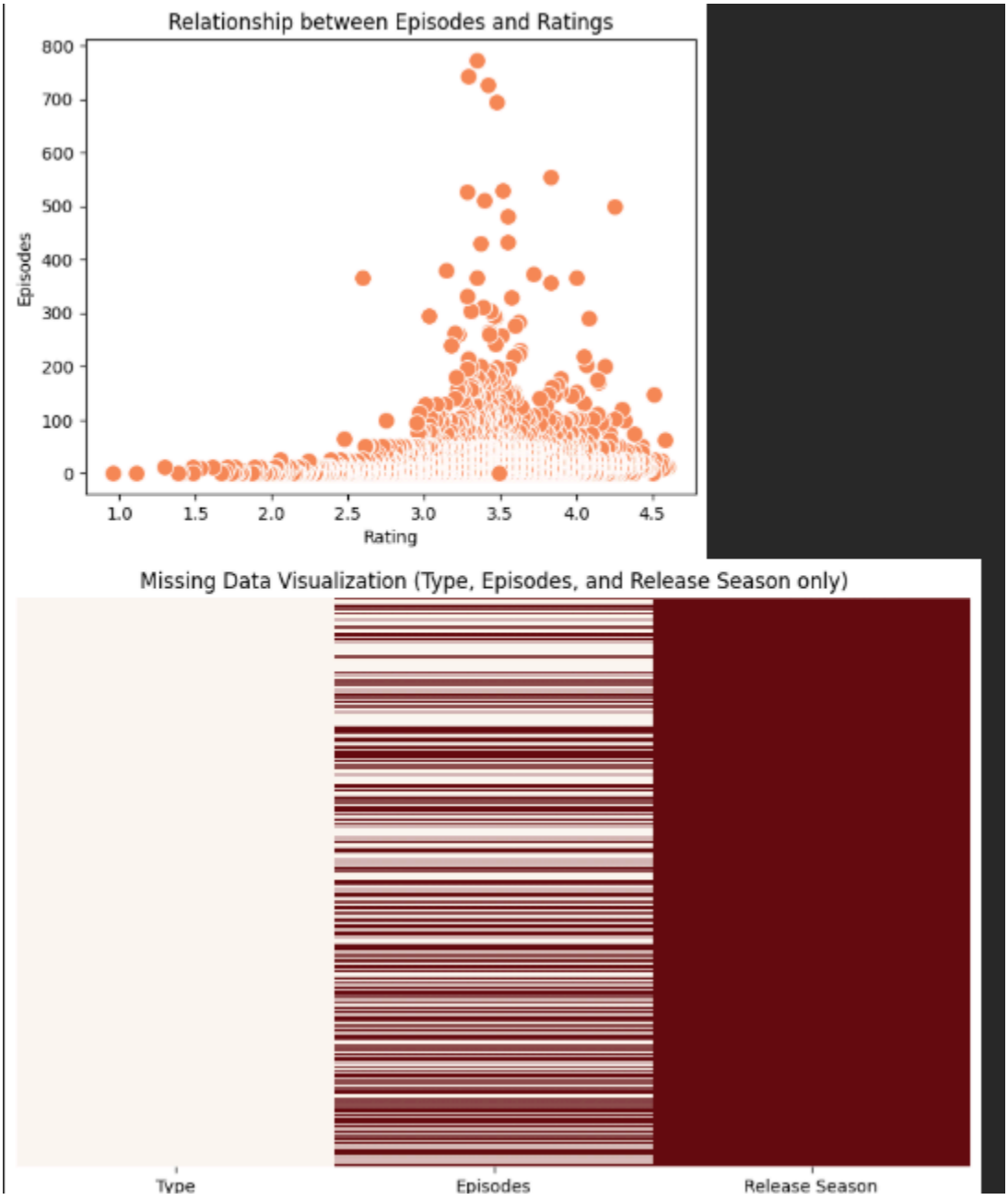
plt.figure(figsize=(10, 6))
sns.heatmap(subset_df.isnull(), cbar=False, cmap="Reds", yticklabels=False)
plt.title("Missing Data Visualization (Type, Episodes, and Release Season only)")
plt.show()

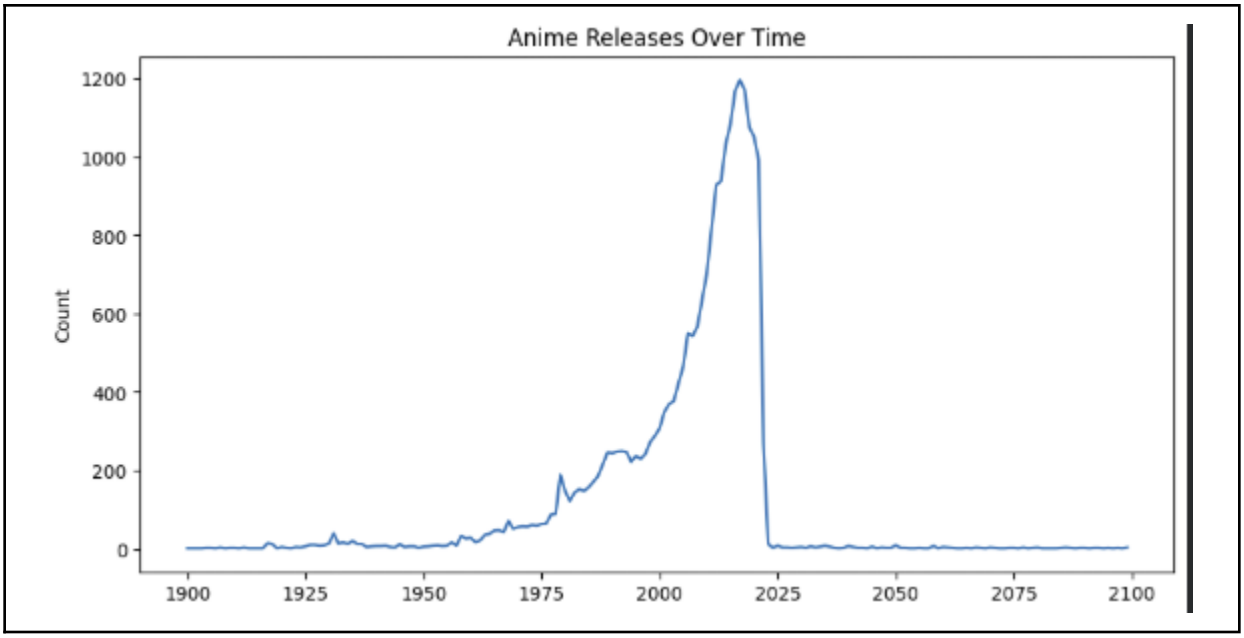
#timeseries
years = df.astype(str).stack().str.extract(r'(\b19\d\d|\b20\d\d)')[0].dropna().astype(int)
years.value_counts().sort_index().plot(kind='line', figsize=(10, 5))

plt.title("Anime Releases Over Time")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```

OUTPUT


Frequency of Media Type
count
5000
4000
3000
2000
1000
0
TV
Web
Movie
OVA
Music
TV Sp
DVD S
Other
Type
Distribution of Ratings
Frequency
4000
3500
3000
2500
2000
1500
1000
500
0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Ratings


Relationship between Episodes and Ratings
Episodes
Rating
800
700
600
500
400
300
200
100
0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Missing Data Visualization (Type, Episodes, and Release Season only)
Type
Episodes
Release Season



ANALYSIS

Univariate analysis
- Categorical: In the Anime franchise, TV series dominated by a wide margin with movies and web coming in second and third respectively.

- Quantitative: The graph shows that the ratings are fair and spread out with an average of around 3.0 to 3.5.

Bivariate analysis
- Scatterplot: Comparing this to the previous graphs, it is quite similar in shape to the quantitative histogram. We can safely say that the episode count does not correlate to a higher rating, rather it's a completely independent variable.

Missing data visualization: I've analyzed that there are 1 type of where the anime can be watched so it is one only. Also, there are a lot of release seasons in every anime. However, there is always a missing episodes in every anime which when the anime is still ongoing.

Time-series: The graph shows the history of anime release throughout the year, starting to 1900s, maintaining low and steady and it starting to go up once it enter the 2000s era.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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